| | $\mathcal{K}_{0^+}^{\#1}$ | $\mathcal{K}_{0^-}^{\#1\,\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{K}_{0^+}^{\#1}\,\dagger$ | $\frac{1}{2}\,(-3\,\Upsilon_1\,k^2 - 3\,\overset{(2)}{\kappa}_3)$ | $0$ |
| $\mathcal{K}_{0^-}^{\#1}\,\dagger^{\alpha\beta\chi}$ | $0$ | $3\,k^2\,\overset{(4)}{\kappa}_{15}$ |

| | $\mathcal{K}_{1^+}^{\#1\,\alpha\beta}$ | $\mathcal{K}_{1^+}^{\#2\,\alpha\beta}$ | $\mathcal{K}_{1^-}^{\#1\,\alpha}$ | $\mathcal{K}_{1^-}^{\#2\,\alpha}$ |
|---|---|---|---|---|
| $\mathcal{K}_{1^+}^{\#1}\,\dagger^{\alpha\beta}$ | $0$ | $0$ | $0$ | $0$ |
| $\mathcal{K}_{1^+}^{\#2}\,\dagger^{\alpha\beta}$ | $0$ | $0$ | $0$ | $0$ |
| $\mathcal{K}_{1^-}^{\#1}\,\dagger^{\alpha}$ | $0$ | $0$ | $-k^2\,\overset{(4)}{\kappa}_2 - \overset{(2)}{\kappa}_3$ | $\frac{k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_3}{\sqrt{2}}$ |
| $\mathcal{K}_{1^-}^{\#2}\,\dagger^{\alpha}$ | $0$ | $0$ | $\frac{k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_3}{\sqrt{2}}$ | $\frac{1}{2}\,(-k^2\,\overset{(4)}{\kappa}_2 - \overset{(2)}{\kappa}_3)$ |

| | $\mathcal{K}_{2^+}^{\#1\,\alpha\beta}$ | $\mathcal{K}_{2^-}^{\#1\,\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{K}_{2^+}^{\#1}\,\dagger^{\alpha\beta}$ | $0$ | $0$ |
| $\mathcal{K}_{2^-}^{\#1}\,\dagger^{\alpha\beta\chi}$ | $0$ | $0$ |

| | $\mathcal{J}_{0^+}^{\#1}$ | $\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{J}_{0^+}^{\#1}\,\dagger$ | $\frac{1}{\text{Det}\,(0^+)}$ | $0$ |
| $\mathcal{J}_{0^-}^{\#1}\,\dagger^{\alpha\beta\chi}$ | $0$ | $\frac{1}{3\,k^2\,\overset{(4)}{\kappa}_{15}}$ |

| | $\mathcal{J}_{1^+}^{\#1\,\alpha\beta}$ | $\mathcal{J}_{1^+}^{\#2\,\alpha\beta}$ | $\mathcal{J}_{1^-}^{\#1\,\alpha}$ | $\mathcal{J}_{1^-}^{\#2\,\alpha}$ |
|---|---|---|---|---|
| $\mathcal{J}_{1^+}^{\#1}\,\dagger^{\alpha\beta}$ | $0$ | $0$ | $0$ | $0$ |
| $\mathcal{J}_{1^+}^{\#2}\,\dagger^{\alpha\beta}$ | $0$ | $0$ | $0$ | $0$ |
| $\mathcal{J}_{1^-}^{\#1}\,\dagger^{\alpha}$ | $0$ | $0$ | $\frac{2}{3\,\text{Det}\,(1^-)}$ | $-\frac{\sqrt{2}}{3\,\text{Det}\,(1^-)}$ |
| $\mathcal{J}_{1^-}^{\#2}\,\dagger^{\alpha}$ | $0$ | $0$ | $-\frac{\sqrt{2}}{3\,\text{Det}\,(1^-)}$ | $\frac{1}{3\,\text{Det}\,(1^-)}$ |

| | $\mathcal{J}_{2^+}^{\#1\,\alpha\beta}$ | $\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi}$ |
|---|---|---|
| $\mathcal{J}_{2^+}^{\#1}\,\dagger^{\alpha\beta}$ | $0$ | $0$ |
| $\mathcal{J}_{2^-}^{\#1}\,\dagger^{\alpha\beta\chi}$ | $0$ | $0$ |

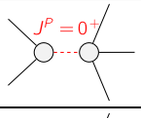
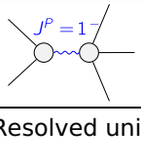
Abbreviations used in matrices

$$\Upsilon_1 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2 \;\&\&\; \text{Det}\,(0^+) == -\frac{3}{2}\,(k^2\,(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2) + \overset{(2)}{\kappa}_3) \;\&\&\; \text{Det}\,(1^-) == -\frac{3}{2}\,(k^2\,\overset{(4)}{\kappa}_2 + \overset{(2)}{\kappa}_3)$$

Lagrangian

$$\overset{(2)}{\kappa}_3\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}^{\chi}{}_{\beta\chi} + \overset{(4)}{\kappa}_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + 2\,\overset{(4)}{\kappa}_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} - 4\,\overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} -$$
$$2\,\overset{(4)}{\kappa}_{15}\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - \overset{(4)}{\kappa}_{15}\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\alpha\beta\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi} - 2\,\overset{(4)}{\kappa}_{15}\,\partial_\delta\mathcal{K}_{\beta\alpha\chi}\,\partial^\delta\mathcal{K}^{\alpha\beta\chi}$$

| Added source term(s): | $\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$ | |
|---|---|---|
| Source constraint(s) | # constraint(s) | Covariant form |
| $\mathcal{J}_{1^-}^{\#1\,\alpha} + 2\;\mathcal{J}_{1^-}^{\#2\,\alpha} == 0$ | 3 | $\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 3\,\partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$ |
| $\mathcal{J}_{1^+}^{\#2\,\alpha\beta} == 0$ | 3 | $\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}$ |
| $\mathcal{J}_{1^+}^{\#1\,\alpha\beta} == 0$ | 3 | $\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$ |
| $\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi} == 0$ | 5 | $3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\;\eta^{\beta\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\;\eta^{\alpha\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\;\eta^{\beta\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\;\eta^{\alpha\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\;\eta^{\beta\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\;\eta^{\alpha\chi}\;\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$ |
| $\mathcal{J}_{2^+}^{\#1\,\alpha\beta} == 0$ | 5 | $3\,\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 3\,\partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\;\eta^{\alpha\beta}\;\partial_\epsilon\partial^\epsilon\partial_\delta\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} == 2\,\partial_\delta\partial^\beta\partial^\alpha\mathcal{J}^{\chi}{}_{\chi}{}^{\delta} + 3\,(\partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi})$ |
| Total # constraint(s): | 19 | |

| Resolved pole(s) | # polarization(s) | Square mass | Residue |
|---|---|---|---|
|   | 1 | $-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2}$ | $-\frac{2}{3\,(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2)}$ |
|   | 3 | $-\frac{\overset{(2)}{\kappa}_3}{\overset{(4)}{\kappa}_2}$ | $\frac{2}{3\,\overset{(4)}{\kappa}_2}$ |
| Resolved unitarity condition(s): | | (Demonstrably impossible) | |